THE UPPER SHRIEK FUNCTORS COMPOSE UP TO A CANONICAL ISOMORPHISM ON BOUNDED BELOW COMPLEXES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The base change isomorphism for the upper shriek of a surjection of finite type algebras over a Noetherian ring. Invoked inside the counted proof (source lines 4794-4796). Identifies the shriek of the polynomial extension with the tensor of the shriek after base change in the bounded below case.
- The identification of the upper shriek of a composition with the composite of the upper shrieks. Invoked inside the counted proof (source lines 4797-4799). Identifies the two ways of writing the shriek of the composite surjection through adjointness to restriction.
- The double base change isomorphism for the shriek of finite type homomorphisms. Invoked at the end of the counted proof (source lines 4811-4812). Identifies the base changed shriek term with the shriek computed over the source.

Source reference. Original lemma lemma-flat-bc-surjection: sources/stacks-dualizing/pinned-original-source.tex, source lines 2726–2726. Navigation pointer only: the counted text refers to the Stacks item lemma-flat-bc-surjection, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference more-algebra-lemma-double-base-change: sources/stacks-dualizing/pinned-original-source.tex, source lines 4695–4695. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-double-base-change, which belongs to another chapter of the project and is not part of this

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Source reference. Original section section-trivial: sources/stacks-dualizing/pinned-original-source.tex, source lines 2416–2416. Navigation pointer only: the counted text refers to the Stacks item section-trivial, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $A \xrightarrow{a} B \xrightarrow{b} C$ be finite type homomorphisms of Noetherian rings. Then there is a transformation of functors $b^! \circ a^! \to (b \circ a)^!$ which is an isomorphism on $D^+(A)$.*

Proof. Choose a polynomial ring $P = A[x_1, \ldots, x_n]$ over A and a surjection $P \to B$. Choose elements $c_1, \ldots, c_m \in C$ generating C over B. Set $Q = P[y_1, \ldots, y_m]$ and denote $Q' = Q \otimes_P B = B[y_1, \ldots, y_m]$. Let $\chi : Q' \to C$ be the surjection sending y_j to c_j. Picture

$$\begin{array}{ccccc} & & Q & \xrightarrow[\psi']{} & Q' \xrightarrow{\chi} C \\ & & \uparrow & & \uparrow \\ A & \longrightarrow & P & \xrightarrow{\psi} & B \end{array}$$

By Lemma lemma-flat-bc-surjection for $M \in D(P)$ we have an arrow $\psi^!(M) \otimes^{\mathbf{L}}_B Q' \to (\psi')^!(M \otimes^{\mathbf{L}}_P Q)$ which is an isomorphism whenever M is bounded below. Also we have $\chi^! \circ (\psi')^! = (\chi \circ \psi')^!$ as both functors are adjoint to the restriction functor $D(C) \to D(Q)$ by Section section-trivial. Then we see

$$\begin{aligned} b^!(a^!(K)) &= \chi^!(\psi^!(K \otimes^{\mathbf{L}}_A P)[n] \otimes^{\mathbf{L}}_B Q)[m] \\ &\to \chi^!((\psi')^!(K \otimes^{\mathbf{L}}_A P \otimes^{\mathbf{L}}_P Q))[n+m] \\ &= (\chi \circ \psi')^!(K \otimes^{\mathbf{L}}_A Q)[n+m] \\ &= (b \circ a)^!(K) \end{aligned}$$

where we have used in addition to the above More on Algebra, Lemma more-algebra-lemma-double-base-change. □